

# Avi Garg

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## EDUCATION

### Thapar Institute of Engineering and Technology

B.E. in Electronics and Communication Engineering (CGPA: 8.79)

Patiala, Punjab

Aug 2024 – May 2028

## EXPERIENCE

### S&T Consultancy Services

June 2026 – Present

Software Engineering Intern

Remote

- Built and maintained 8+ RESTful API endpoints in **Node.js** and **Express.js**, supporting core functionality for production applications used by internal and client-facing teams.
- Designed and normalized SQL/NoSQL database schemas and optimized queries across **PostgreSQL** and **MongoDB**, reducing average query response time by 20% and improving application performance.
- Debugged and resolved 3+ production issues per sprint, wrote unit and integration tests, and collaborated in Agile ceremonies (stand-ups, sprint planning) to ship features on schedule.
- Participated in code reviews and CI/CD deployment pipelines, improving code quality and reducing release-cycle defects.

## PROJECTS

### AI Customer Support Platform

Personal Project

Next.js, TypeScript, PostgreSQL, ScaleKit, OpenAI API

2026

- Built a full-stack, multi-tenant AI customer support platform in **Next.js** and **TypeScript** that automates customer interactions using the **OpenAI API** for conversational AI.
- Implemented secure authentication, organization management, and role-based access control with **ScaleKit**, enabling multiple businesses to independently manage isolated chatbot instances.
- Architected a **PostgreSQL** schema to persist business profiles, chatbot configurations, conversation history, and knowledge-base content at scale.

### AI-Powered Delivery Failure Prediction System

Hackathon Project

Python, Jupyter Notebook, React, FastAPI, Express, Vercel, Render

2026

- Trained a machine learning model on **Amazon delivery data** to predict at-risk orders, engineering a **61-feature** matrix including custom *traffic-pressure*, *reliability-risk*, and *distance-cost* metrics.
- Tuned an **XGBoost Classifier** using **5-fold cross-validation**, achieving **96.18%** validation accuracy, a **0.93** F1-score, and a **0.99** AUC-ROC.
- Integrated a React.js frontend with a FastAPI backend via REST APIs for real-time prediction results, and deployed the system on Vercel and Render.

### Prodivity Technologies – Full Stack

Freelance Agency Platform

React.js, Node.js, Express.js, MongoDB, Tailwind CSS

2026

- Co-developed a full-stack **freelance agency platform** centralizing developer portfolios and **client acquisition** for a collective of freelance developers, using React.js, Node.js, and MongoDB.
- Built a secure **admin dashboard** allowing non-technical team members to create and update projects and services dynamically, without code changes.
- Designed RESTful, **MongoDB**-backed APIs for content management and real-time frontend-backend data synchronization.

## TECHNICAL SKILLS

**Languages:** C++, JavaScript, TypeScript, SQL, Python

**Frameworks & Libraries:** Next.js, Langchain, React.js, Node.js, Express.js, FastAPI, Tailwind CSS

**Databases:** PostgreSQL, MySQL, MongoDB (NoSQL)

**Cloud & DevOps:** AWS, Docker, Render, Vercel, CI/CD, Git, GitHub, Postman

**Core CS Concepts:** Data Structures & Algorithms (DSA), Operating Systems (OS), Computer Networks (CN), Distributed Systems, REST API Design, Unit Testing, Agile/Scrum

**Machine Learning:** XGBoost, Scikit-learn, Feature Engineering, Model Evaluation

## ACHIEVEMENTS

- Open Source & Version Control:** Managed and deployed **3+** full-stack production repositories on GitHub, applying advanced Git branching strategies and CI/CD workflows.
- Shortlisted at **Hack Helix 2026** for the AI-Powered Delivery Failure Prediction System, an ML-based logistics risk model.
- Participated in the **GDG Solution Challenge** (Hack2Skill), building solutions aligned with Google Developer Groups' problem statements.